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	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 22

A. (1) Research Program/Project/Study Title:	
HIV Awareness and Preventive Measures among Tertiary Students of CapSU: Basis for Developing Supplementary Materials for English, Science, and Mathematics	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:							
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								

	Document Type:	Document Code	RES-F08
		Revision No.	00
	Document Title:	Effective Date	January 26, 2023
		Page	Page 2 of 22
(7) Sector/Commodity		(8) Discipline/Field of Specialization	

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 3 of 22

(9A) Abstract

With the prevalence of HIV and AIDS in the Philippines, the country is one of only seven countries globally where the number of new HIV cases has increased by over 25 per cent from 2001 to 2009. It is on this premise why this study was formulated. This research aimed to determine the level of HIV awareness and preventive measures among tertiary students of Capiz State University. Moreover, instructional materials on English, Science and Mathematics are developed based on the results of this research. This study used a causal comparative correlation research design to identify the level of HIV awareness and preventive measures among tertiary learners of CapSU and to established the relationships among variables of the study. Results showed that tertiary students are aware of the issue on HIV trends and HIV transmission. Also, tertiary students are aware of the HIV preventive measures like HIV testing, risk reduction, treatment and diagnosis and high-risk situations. Further, when grouped according to their demographic profiles, it was revealed that no significant difference existed on their HIV awareness and HIV preventive measures when categorized according to their profiles. Meanwhile, results further presented a strong relationship between HIV awareness and HIV preventive measures. This suggests that an increase in the tertiary learners' HIV awareness will have immense chance of increasing their level of their HIV preventive measures. Finally, based on the results of the study a supplementary material on English, Science and Mathematics were crafted in the hope that these materials will further increase students' awareness on HIV, thereby, increasing their knowledge on HIV preventive measures.

Keywords

sexually transmitted infections, tertiary students, supplementary materials, virus

(10) Introduction

Human immunodeficiency virus (HIV) is one of the many complex health problems in the world. Young people are at high risk of HIV infections and are therefore, in need of target prevention. In the past decade, the Philippines has gained notoriety as the country with the fastest-growing human immunodeficiency virus (HIV) epidemic in the Western Pacific region (Ganguangco, 2019) While the overall trend of HIV incidence and AIDS-related deaths are declining globally (GBD 2017 HIV Collaborators, 2017), an increase in new cases was reported to the HIV/AIDS and ART Registry of the Philippines (HARP) in the recent decade as reflected in figure 1.

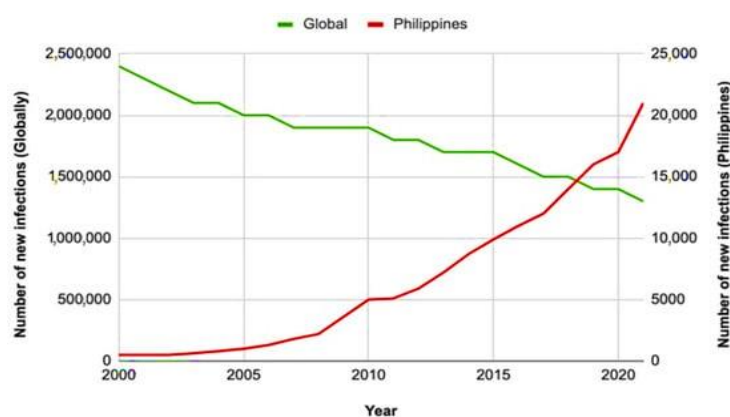


Figure 1. Estimated annual new HIV infections among individuals 15 years old and above from 2000–2021 globally and in the Philippines, based on UNAIDS estimates.

(Source: Ganguangco and Eustaquio, 2023)

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 22

In 2012, there were only approximately nine new HIV cases every day. In 2023, however, there have been 46 cases reported daily, a stunning 411% increase in daily incidence in 10 years. Late presentation in care remains a concern, with 29% of new confirmed HIV cases in January 2023 having clinical manifestations of advanced HIV disease at the time of diagnosis (DOH, 2023)

DOH Capiz reported that 89% of the increase in HIV occurrence in Capiz and is mostly male victims. Since the virus knows no age and gender, status and income, it has become a threat especially to the young individuals. With these alarming cases all over the Philippines, it is not far that one day, the cases of HIV victims in the province will rise, too. Considering this possibility, awareness campaigns have been done and are still continuously conducted by the proponents, advocates and health organizations. Alhasawi, et.al (2019) stated that in many countries, a significant proportion of youth initiate sexual activity by the age of 15 years. Thus, adolescents in general are at higher risk of contracting HIV. It is in this premise that this study was conducted.

Generally, this research aimed to identify the level of HIV awareness and preventive measures among tertiary students of Capiz State University. Results gathered served as a springboard in developing supplementary materials for English, Science, and Mathematics. Specifically, this study would like to answer the following questions: What is the profile of the respondents according to sex, parents' educational status and occupation, place of residence, and type of school? What is the level of HIV awareness among tertiary students of Capiz State University? What is the level of perceived preventive measures among tertiary students of Capiz State University? Is there a significant difference in the level of HIV awareness among tertiary students of Capiz State University when classified according to age, sex, parents' educational status and occupation and place of residence? Is there a significant difference in the level of HIV preventive measures among tertiary students of Capiz State University when classified according to sex, parents' educational status and occupation, place of residence? Is there a significant relationship between the respondents' HIV awareness and preventive measures? What supplementary materials in English, Science, and Mathematics may be developed from the results of the study?

With the results of the study, this will give the university a birds-eye-view of the scenario of the teenagers, which are widely the clientele of the University itself. Therefore, this could serve as a guide in planning and conducting significant activities on awareness campaign and preventive measures, symposia, and the like. This study might serve as an eye-opener to every higher education learner to every possibility and possible consequences of their irresponsible actions. This will assess themselves if how aware they are on this problem the society faces and on how not to become a victim of the virus. Also, with the developed supplementary materials, aside from being used in the classroom, this could set as a reminder and outlet of information about issues on HIV.

Further, this study was guided by the following framework:

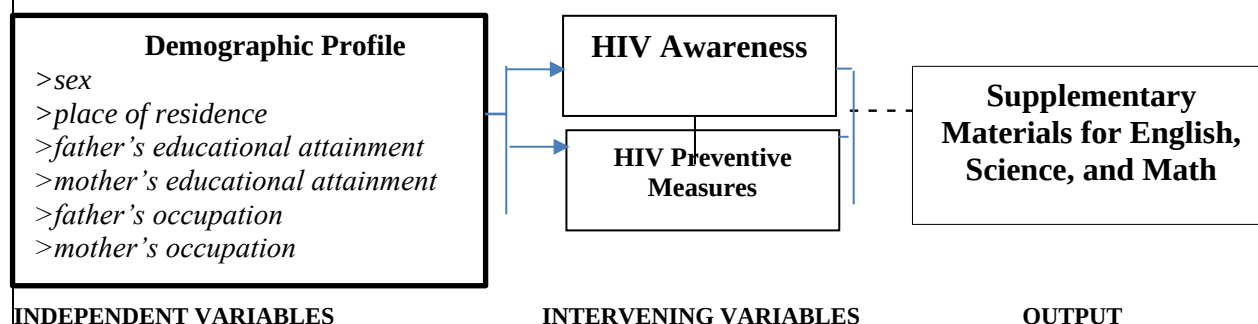


Fig. 1. The conceptual framework of the study.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 22

(11) Materials and Methods or Methodology

Research Design

This study was conducted using quantitative research, specifically a causal-comparative correlation research. A causal- comparative method is used to identify the cause-and-effect relationships between a dependent and independent variable. In a causal comparative research design, the researcher compares two groups to find out whether the independent variables affected the outcome of the dependent variable (Graham, 2022). Meanwhile, correlational research was also used in this study. Creswell (2012) states that correlation research, describe and measure the degree of association (or relationship) between two or more variables.

Research Instruments

The study utilized a validated and pilot tested researcher-made checklist to identify the level of consciousness and preventive measures of the respondents. The survey checklist was validated by poll of experts in the field of science such as the science teachers and professors and an expert from the office of the population commission in the province of Capiz. Moreover, researcher-made checklist was able to achieved a Cronbach alpha level of 0.88. Cronbach alpha measures the internal consistency or reliability of a set of survey items. A value of 0.7 or higher is already an acceptable survey instrument (Frost, 2023).

The research instrument is divided into three parts, namely; part I asked for respondents' demographic profile as to sex, place of residence, parents' educational attainment and parents' occupation. Part II was consisted of HIV awareness in terms of HIV trends and transmission. Part III inquired about the respondents' perceived preventive measures which was categorized into HIV testing, risk reduction, treatment and diagnosis and high-risk reduction.

Respondents

The respondents of the study were the tertiary students of Capiz State University who were officially enrolled during the academic year 2021-2022. This study included 377 tertiary students of CapSU out of 21, 182 who were officially enrolled. Stratified random sampling was used in determining the sample size from each campus.

Data Gathering Procedure

This study was conducted during academic year 2021-2022 when classes were conducted in a hybrid (both online and limited face to face) sessions. Research questionnaires were sent out to the respondents of this study using the Google form to each of the nine campuses of this university. Data collection lasted for 60 days to finally come up with the appropriate number of respondents per campus. When the needed data were gathered, respondents' level of HIV awareness and preventive measures were then interpreted using the following scales:

Mean	Verbal Interpretation	Descriptions
4.50 – 5.00	Fully Aware	Students are equipped with all the details/ knowledge of the topic or issue presented.
3.50 – 4.49	Aware	Students are equipped with information

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 22
2.50 – 3.49 Moderately Aware about the topic or issue presented but needs to be reinforced. Students have limited information about the topic or issue presented. 1.50 – 2.49 Poorly Aware Students are informed but with some misconceptions about the topic or issue presented. 1.00 – 1.49 Unaware Students have no knowledge at all about the topic or the issue presented. Further for the analysis of data gathered, both descriptive statistics and inferential statistics were used. Means and percentages were utilized for descriptive analysis. While, Mann-Whitney U test, Kruskal Wallis test and Spearman's rho were used for inferential statistics.				
(12) Results and Discussions RESULTS Table 1 presents the demographic profile of the tertiary students of CAPSU. Data revealed that most of the respondents who participated in the survey are female (298 or 79%) and only few are male (79 or 21%). With regards to the father's educational attainment, out of 377 respondents, 132 or 35% of the respondents' father are high school graduate and 245 or 65% are at most high school level. Also, for the mother's educational attainment, 199 or 52.79% are high school graduate and 178 or 47.21% are at most high school level. Furthermore, data revealed that most of the father's occupation, are labor jobs 324 or 85.94%, followed by non-professional jobs 30 or 7.96%, 14 or 3.71% have been accounted for business related jobs and only 9 or 2.38% have professional jobs. Mother's occupation are mostly housewives with 296 or 78.51%, followed by labor jobs which is 48 or 12.73%, 19 or 5.04% are non-professional jobs and only 14 or 3.71% have professional jobs. Also, there are 173 or 45.89% respondents who live in urban areas and 204 or 54.11% live in rural areas. This means that most of the respondents are female, majority of their father's educational attainment is on high school level and that of the mother's educational attainment is high school graduate. Furthermore, most of the father's occupation is labor jobs and most of the mother's job is housewife. Majority of the respondents reside in rural area. Table 2 presents the level of HIV Consciousness of the tertiary students of CAPSU. Generally, data revealed that the level of HIV consciousness of the respondents is "Aware" ($M=3.94, SD= .503$). Specifically, data revealed that the respondents are also "Aware" in the level of HIV awareness in terms of HIV Trends ($M=4.0, SD= .575$) and HIV Transmission ($M=3.82, SD=.661$). This means that the students are equipped with information about the different trends on HIV and its transmission but needs to be reinforced. Table 3 presents the level of perceived HIV preventive measures. Generally, data revealed that the level of perceived HIV preventive measures is "Aware" ($M=3.81, SD=.833$). Specifically, data revealed that the highest mean falls in terms of Testing ($M= 3.86, SD= .803$), followed by High Risk Situations ($M=3.85, SD= .999$), Risk Reduction ($M= 3.73, SD= .957$) and Treatment and Diagnosis being the lowest ($M=3.72, SD=.905$). This implies that students are equipped with information about the different HIV preventive measures but still needs to be reinforced. Although the students claim to be conscious about HIV preventive measures, true understanding of the disease prevention is still lacking.				

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 7 of 22

Table 4 presents the significant difference between HIV awareness and the respondents' demographic profile. Data revealed that no significant difference exists on the respondents' HIV awareness when classified in terms of their profiles, sex ($p=.496$), place of residence ($p=.496$), father's educational attainment ($p=.247$), mother's educational attainment ($p=.646$), father's occupation ($p=.587$) and mother's occupation ($p=.270$).

Table 5 presents the significant difference on HIV preventive measures in terms of respondents' demographic profile. Data revealed that no significant difference exists in respondents' HIV preventive measures when grouped according to sex ($p=.514$), place of residence ($p=.536$), father's educational attainment ($p=.219$), mother's educational attainment ($p=.267$), father's occupation ($p=.891$) and mother's occupation ($p=.355$).

Table 6 shows the relationship between HIV awareness and HIV preventive measures. It was found out that there is a strong, positive relationship between HIV awareness and HIV preventive measures among tertiary learners of Capiz State University.

DISCUSSION

Tertiary learners of Capiz State University, regardless of sex, place of residence, parents' educational attainment, parents' occupation, revealed that they are all aware of HIV issues such as on HIV trends and HIV transmission. Lessons in science, specifically in biology, includes topics like reproductive system wherein, at the latter part, touches on sexually-transmitted diseases. This gives the learners prior knowledge about the topics. Moreover, learners are now bombarded with information from various social media platforms about HIV – its basic information and how it can be transmitted from one person to another. As mentioned by Srivasta, et. al (2021), with the increase in age and years in schooling, the HIV awareness increased among boys and girls. Also, use of internet among adolescent boys and girls was positively associated with HIV awareness. Choruma et.al. (2021) stated in his study that most students obtained their information on HIV and AIDS from television and radio programs. Just like what Nabisubi (2021) stated that students' major source on HIV are the teachers in school. Another that can account for the increased in awareness of learners about the topic is increasing efforts from the government and other non-government organizations to make everyone in the community become conscious of the presence of this kind of virus, thus, making information about them readily and easily accessible to everyone. On the other hand, the respondents' awareness on HIV preventive measures could also be explained by the reasons mentioned above.

There is no significant difference in the HIV awareness and preventive measures when categorized based on respondents' demographic profile. The "aware" level of awareness and prevention measures of respondents means that students are equipped with information about the topic but could be reinforced. As mentioned by Alhasawi, et.al. (2019) their study that revealed that the students were knowledgeable with regard to the nature and mode of transmission of HIV/AIDS, but they needed a more detailed understanding of the disease to prevent stigmatization and discrimination of an infected person. This claim was supported by Nabisubi et.al. (2021) when she reported that despite the high level of awareness on HIV/AIDS of young people aging from 12 – 20, aspects like cause and modern prevention methods like taking prep and prevention of mother to child transmission were less known to them. Which made her suggested a strong emphasis on up-to-date control and prevention methods should be implemented to fight the HIV/AIDS scourge. In the case of tertiary students of Capiz State University their awareness of the topic or issue presented to them may be enhanced through listening from various digital and non-digital

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 8 of 22

materials, campaigns, news articles and the like. As suggested by Gao, et.al (2012) HIV/AIDS education programs positively influenced HIV/AIDS related knowledge and attitudes. A systematic and long-term intervention among students must be conducted for the prevention of HIV.

Results of this study revealed that tertiary students are only “aware” of HIV awareness and preventive measures. However, it was also exposed that there is a strong and positive relationship between HIV awareness and HIV preventive measures which simply displayed that once the tertiary learners’ HIV awareness is increased, ultimately their level of preventive measures will also be increased. These outcomes gave good opportunity to come up with a plan on how to make the tertiary learners become fully aware of the recent trends and needs on HIV prevention and control. With this, supplementary materials in English, Science and Mathematics were produced. These materials all aimed at incorporating topics like basic trends in HIV, transmission, HIV treatment and HIV risk reduction in English, Science and Mathematics lessons to expand students’ consciousness of the said topics.

Education is a life-long process and one of its aims is to produce learners who are responsible and productive members of society. It lies in the hands of the teachers on how these learners achieve what is expected from them. Through these supplementary materials produced as an output of this study, it is with great anticipation that these supplementary materials could serve as bridge in improving the learners’ awareness on HIV, increasing their preventive measures which will finally lead to a decrease in the number of HIV reactants in the community.

(13) Conclusions

From the different findings of the study, it can be concluded that tertiary learners of Capiz State University are only aware of the HIV trends and transmission, HIV testing, risk reduction, treatment and diagnosis and high-risk reduction. There is a need to upgrade the level of awareness for them to become fully aware so they may be equipped with all the details, and even may become advanced learner, on the perils of having or inhabiting the virus. Meanwhile, the strong, positive relationship between HIV awareness and HIV preventive measures is well established in this study. Greater awareness of HIV and on its modes of transmission leads to an increased likelihood of individuals adopting preventive measures. When people are informed, they are more likely to take actions that would prevent them from contacting the disease. Therefore, promoting HIV awareness remains a crucial component of HIV prevention strategies.

(14) Output

The output of this study were supplementary materials in subjects such as English, Science and Mathematics. Since it was established that there is a strong, positive relationship between the respondents’ HIV awareness and preventive measures, it was decided that a supplementary material focusing on HIV awareness must be crafted.

(please see attached sample outputs of the study)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 9 of 22

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 10 of 22

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 11 of 22

Appendices

Table 1. Profile of respondents

Profile	Frequency	Percentage
Sex at birth		
Male	79	21%
Female	298	79%
Father's Educational Attainment		
At least HS graduate	132	35%
At most HS graduate	245	65%
Mother's Educational Attainment		
At least HS graduate	199	52.79%
At most HS level	178	47.21%
Father's Occupation		
Professional jobs	9	2.38%
Non-professional jobs	30	7.96%
Business related jobs	14	3.71%
Labor jobs	324	85.94%
Mother's Occupation		
Professional jobs	14	3.71%
Non-professional jobs	19	5.04%
Business related jobs	0	0
Labor jobs	48	12.73%
N/A – Housewife	296	78.51%
Place of Residence		
Urban	173	45.89%
Rural	204	54.11%

Table 2. Level of HIV awareness

	MEAN	STANDARD DEVIATION	VERBAL INTERPRETATION
HIV Awareness			
HIV Trends	4.00	.575	Aware
HIV Transmission	3.82	.661	Aware
Over all	3.94	.503	Aware

Table 3. Level of perceived HIV preventive measures

HIV Preventive Measures	MEAN	STANDARD DEVIATION	VERBAL INTERPRETATION
Testing	3.86	.803	Aware
Risk Reduction	3.73	.957	Aware
Treatment and Diagnosis	3.72	.905	Aware
High Risk Situations	3.85	.999	Aware
Over all	3.81	.833	Aware

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 12 of 22

Table 4. Significant difference on HIV awareness in terms of respondents' sex, place of residence, parents' educational attainment and parents' occupation,

Respondents' Profile	Test statistics	t-value	p-value
Sex	Mann-Whitney U	-.680	.496
Place of Residence	Mann-Whitney U	-.680	.496
Father's Educational Attainment	Kruskall Wallis	6.664	.247
Mother's Educational Attainment	Kruskall Wallis	4.228	.646
Father's Occupation	Kruskall Wallis	7.484	.587
Mother's Occupation	Kruskall Wallis	8.764	.270

Table 5. Significant difference on HIV preventive measures in terms of respondents' profiles/

Respondents' Profile	Test Statistics	t value	p-value
Sex	Mann-Whitney U	-.653	.514
Place of residence	Mann-Whitney U	-.618	.536
Father's Educational Attainment	Kruskall Wallis	7.016	.219
Mother's Educational Attainment	Kruskall Wallis	7.629	.267
Father's Occupation	Kruskall Wallis	4.300	.891
Mother's Occupation	Kruskall Wallis	7.752	.355

Table 6. Significant relationship between HIV awareness and HIV preventive measures

Variable	Test statistics	r	sig. 2-tailed
HIV awareness and HIV Preventive Measures	Spearman's rho	.776	.000**

**** correlation is significant at the 0.01 level (2-tailed)**

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 13 of 22

Sample Outputs

Supplementary Material for English:

SUPPLEMENTARY MATERIAL IN SPEECH AND ORAL COMMUNICATION

TOPIC: ACHIEVING COMMUNICATIVE COMPETENCE IN VARIOUS SPEECH CONTEXTS

Objectives: Learners should have:

1. Engaged in different communication processes on the topic given about the stage of HIV Epidemic in the Philippines;
2. Apply the guidelines and things to consider on becoming an effective and engaged communicator;
3. Reflect on the theme of the article with an essay; and
4. Design an IEC Material with contents on HIV awareness.

The State of the HIV Epidemic in the Philippines: Progress and Challenges in 2023 by Louie Mar A. Gangcuangco and Patrick C. Eustaquio

Abstract: In the past decade, the Philippines has gained notoriety as the country with the fastest growing human immunodeficiency virus (HIV) epidemic in the Western Pacific region. While the overall trends of HIV incidence and acquired immunodeficiency syndrome (AIDS)-related deaths are declining globally, an increase in new cases was reported to the HIV/AIDS and ART Registry of the Philippines. From 2012 to 2023, there was a 411% increase in daily incidence. Late presentation in care remains a concern, with 29% of new confirmed HIV cases in January 2023 having clinical manifestations of advanced HIV disease at the time of diagnosis. Men having sex with men (MSM) are disproportionately affected. Various steps have been taken to address the HIV epidemic in the country. The Philippine HIV and AIDS Policy Act of 2018 (Republic Act 11166) expanded access to HIV testing and treatment. HIV testing now allows for the screening of minors 15–17 years old without parental consent. Community-based organizations have been instrumental in expanding HIV screening to include self-testing and community-based screening. The Philippines moved from centralized HIV diagnosis confirmation by Western blot to a decentralized rapid HIV diagnostic algorithm (rHIVd_a). Dolutegravir-based antiretroviral therapy is now the first line. Pre-exposure prophylaxis in the form of emtricitabine–tenofovir disoproxil fumarate has been rolled out. The number of treatment hubs and primary HIV care facilities continues to increase. Despite these efforts, barriers to ending the HIV epidemic remain, including continued stigma, limited harm reduction services for people who inject drugs, sociocultural factors, and political deterrents. HIV RNA quantification and drug resistance testing are not routinely performed due to associated costs. The high burden of tuberculosis and hepatitis B virus co-infection complicate HIV management. CRF_01AE is now the predominant subtype, which has been associated with poorer clinical outcomes and faster CD4 T-cell decline. The HIV epidemic in the Philippines requires a multisectoral approach and calls for sustained political commitment, community involvement, and continued collaboration among various

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 14 of 22

stakeholders. In this article, we outline the current progress and challenges in curbing the HIV epidemic in the Philippines.

Introduction

As of January 2023, there were 110,736 HIV cases reported in the Philippines. Although this number seems low considering that the country has over 109 million people, the pervasive stigma, sociopolitical conditions, and barriers to healthcare services are fueling the epidemic in marginalized populations. The number of people living with HIV (PLHIV) is projected to increase by 200% from 158,400 in 2022 to 364,000 by 2030. Despite these challenges, there were advances in the rollout of newer antiretroviral agents, access to pre-exposure prophylaxis, and healthcare legislations that positively impact HIV treatment and prevention. In this narrative review, we outline the current progress and challenges in curbing the HIV epidemic in the Philippines.

Populations disproportionately affected by HIV include key populations, comprising 92% of the new infections in 2022, and vulnerable populations. Key populations include males having sex with males (MSM), transgender women, sex workers, trafficked women and girls, and people who inject drugs (PWID). Vulnerable populations include migrant workers, people with disabilities, people in enclosed spaces, and female partners of key populations.

Men Having Sex with Men

Sexual transmission remains to be the predominant mode of HIV acquisition in the Philippines, primarily among MSM. One of the earliest HIV prevalence studies among MSM was conducted in 2010. HIV testing performed outside the entertainment areas/gay bars of Manila found that, among 406 MSM screened using rapid HIV antibody test kits, 48 tested positive (11.8% [95% confidence interval: 8.7% to 15.0%]). Forty participants consented to a Western blot confirmatory test, with 39 participants testing positive for HIV-1 and one patient testing positive for both HIV-1 and HIV-2. Data from the Philippine Department of Health (DOH) in January 2023 showed that approximately 70% of all HIV cases were among males who have sex with other males, and 17% were among males who have sex with both males and females

Persons Who Inject Drugs

HIV transmission through the sharing of infected needles remains relatively low in the Philippines. It was reported to be highest in 2010, accounting for 9% of all new HIV cases that year. In the same year, an outbreak of HIV and hepatitis C virus (HCV) occurred in Cebu City, where over 50% of PWID were found to have HIV, and 93% were infected with HCV. HIV transmission through infected needles has decreased since 2011 and constitutes ~1% of all newly reported cases in the past few years. However, due to the recent sociopolitical climate, particularly during former president Rodrigo Duterte's "war on drugs", it is likely that data among PWID and among those who use illicit drugs remain

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 15 of 22

underreported for fear of legal ramifications. The last biobehavioral surveillance data among PWID were reported in 2015.

Needle and syringe programs (NSPs) provide access to sterile needles and syringes and facilitate their safe disposal. NSPs have been shown to effectively reduce HIV transmission. However, the implementation of NSPs remain difficult in the Philippines given that the Comprehensive Dangerous Drugs Act of 2002 (Republic Act [RA] 9165) considers any unauthorized possession of drug paraphernalia as *prima facie* evidence of self-administration of dangerous drugs. Violation could lead to a maximum of 4 years imprisonment and a minimum monetary penalty of PHP 10,000 (~USD 182) [13], which is higher than the upper bound of the average monthly minimum wage in 2022. Political pressure and the Dangerous Drugs Act of 2022 continue to make NSPs inaccessible, deterring effective community-based comprehensive HIV prevention services among PWID. The Big Cities Project (BCP) in Cebu funded by the World Bank and the Asian Development Bank aimed to reduce HIV transmission by reducing risk behaviors among PWID. However, the sterile needle distribution aspect of this project was halted within 5 months of implementation due to political pressure. Mental health and substance use disorders need to be addressed as part of a comprehensive response to the HIV epidemic in the Philippines.

Transgender Populations

In the HIV surveillance systems of the Philippines, transgender women were previously included under MSM until 2018 when the HARP and the Integrated HIV Behavioral and Serologic Surveillance (IHBS) disaggregated data based on gender identity. Of the total reported new HIV cases in January 2023, 3% were transgender women. Unique concerns among transgender populations in the Philippines include differences in HIV knowledge, need for safe sex communication, and disparate access to healthcare services. One study among trans men and trans women ($N = 525$) in Metro Manila showed that as many as 82% declined HIV testing and counseling services. Amid the gaps in healthcare access, a community-led health service delivery model providing integrated HIV and gender-affirming care showed promise in providing essential health services to transgender people.

Other Vulnerable Groups

There are other populations in the Philippines vulnerable to HIV. These include migrant workers, people who exchange sex, people trafficked for sex, and people in enclosed spaces.

Despite efforts to abolish policies that discriminate against PLHIV, several countries continue to mandate HIV antibody testing prior to migration or employment. Historically, overseas Filipino workers (OFWs) have comprised a significant proportion of HIV cases, likely because of increased case detection from the pre-departure HIV testing imposed by their prospective employers abroad. In a 2006 publication, it was reported that, of 2410 HIV seropositive

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 16 of 22

Activity 1: Dyadic Communication and/or Small Group Discussion

Guide Questions for Discussion:

1. What is your level of knowledge about HIV before and after reading the article?
2. What are your discoveries upon reading the article? Discuss with pair or group.
3. How would you spread awareness regarding HIV among your friends or other people?
4. How would you deal with a person with symptoms of HIV or were confirmed with HIV?

Activity 2: Creating an IEC Material

1. Create an IEC Material to spread awareness on HIV
2. Present the material in class
3. Post on social media accounts the IEC Material made
4. Campaign the sharing and liking of the posted IEC Material to spread awareness

Some References:

Gangcuangco, L. A. & Eustaquio, P. C. 2023. The State of the HIV Epidemic in the Philippines: Progress and Challenges in 2023. *Tropical Medicine and Infectious Disease Journal*.

Gangcuangco, L.M.; Tan, M.L.; Berba, R.P. Prevalence and risk factors for HIV infection among men having sex with men in Metro Manila, Philippines. *Southeast Asian J. Trop. Med. Public Health* **2013**, *44*, 810–817. [Google Scholar]

Philippine Statistics Authority. Age and Sex Distribution in the Philippine Population (2020 Census of Population and Housing). Available online: <https://psa.gov.ph/population-and-housing/node/167965> (accessed on 20 February 2023).

World Health Organization. *Consolidated Guidelines on HIV Prevention, Testing, Treatment, Service Delivery and Monitoring: Recommendations for a Public Health Approach*; World Health Organization: Geneva, Switzerland, 2021. [Google Scholar]

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 17 of 22

Sample Supplementary Material for Mathematics:

Supplemental Materials in **Mathematics in the Modern World**

HIV can be integrated in exponential function:

- **Integration with HIV**
- The short-term (< 4 weeks) viral load response after anti-retroviral therapy can be approximately represented by an exponential decay function of time. The rate of decay is related to the half-life of infected CD4+ T cells;
- The viral load rebound (viral load blips) after termination of therapy can be approximately represented by an exponential growth function of time. The rate of growth is usually greater than the decay rate as in a).
- Graphical and analytical solutions - plot the viral load response and rebound after initiation of therapy and termination of therapy respectively.

As of 2023, it is estimated that the Philippines HIV cases is about 110,736(DOH, 2023). The health advocates and doctors keep on tracking the HIV cases in the country. Including its growth. Mathematics can be used to model the HIV population growth. Recall the formula for exponential growth is $A = Pe^{rt}$, where A is the size of HIV population after it grows. P is the initial number of people with HIV, r is the rate of growth, and t is the time. Recall further e is Euler's constant with n approximate value of 2.718. Plugging in values to this formula-would result in the population size after t time with a growth rate of r.

Example:

The exponential growth model $A = 15e^{(.02)t}$ describes the HIV population of the Manila in thousands, t years after 2010.

- What was the HIV cases in Manila in 2010?
- What will be the HIV population in 2023?

Solution:

- Since our exponential growth model describes the HIV population t years after 2010, we consider 2010 as $t=0$ and then solve for A, our HIV population size.

$$A = 15e^{(.02)t}$$

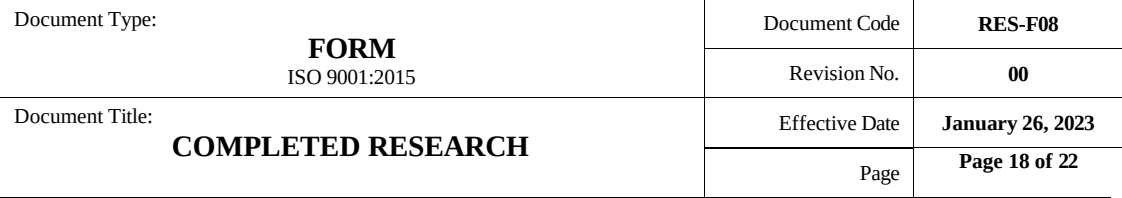
$$A = 15e^{(.02)(0)} \quad \text{Replace } t \text{ with } t=0$$

$$A = 15e^0 \quad e^0 = 1$$

$$A = 15$$

Therefore, the city population in 2010 was 15,000.

- We need to find A for the year 2023. To find t, we subtract 2023 and 2010 to get $t=13$, which we then plug in to our exponential growth model.



Sexual transmission remains to be predominant mode of HIV acquisition in the Philippines, primarily among MSM. One of the earliest HIV prevalence studies among MSM was conducted in 2010. HIV testing performed outside the entertainment areas/gay bars of Manila found that, among 406 MSM screened using rapid HIV antibody test kits, 48 tested positive (11.8% [95% confidence interval: 8.7% to 15.0%]). Forty participants consented to a Western blot confirmatory test, with 39 participants testing positive for HIV-1 and one patient testing positive for both HIV-1 and HIV-2. Data from the Philippine Department of Health in January 2023 showed that approximately 70% of all HIV cases were among males who have sex with other males, and 17% were among males who have sex with both males and females.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 19 of 22

Sample Supplementary Material in Science

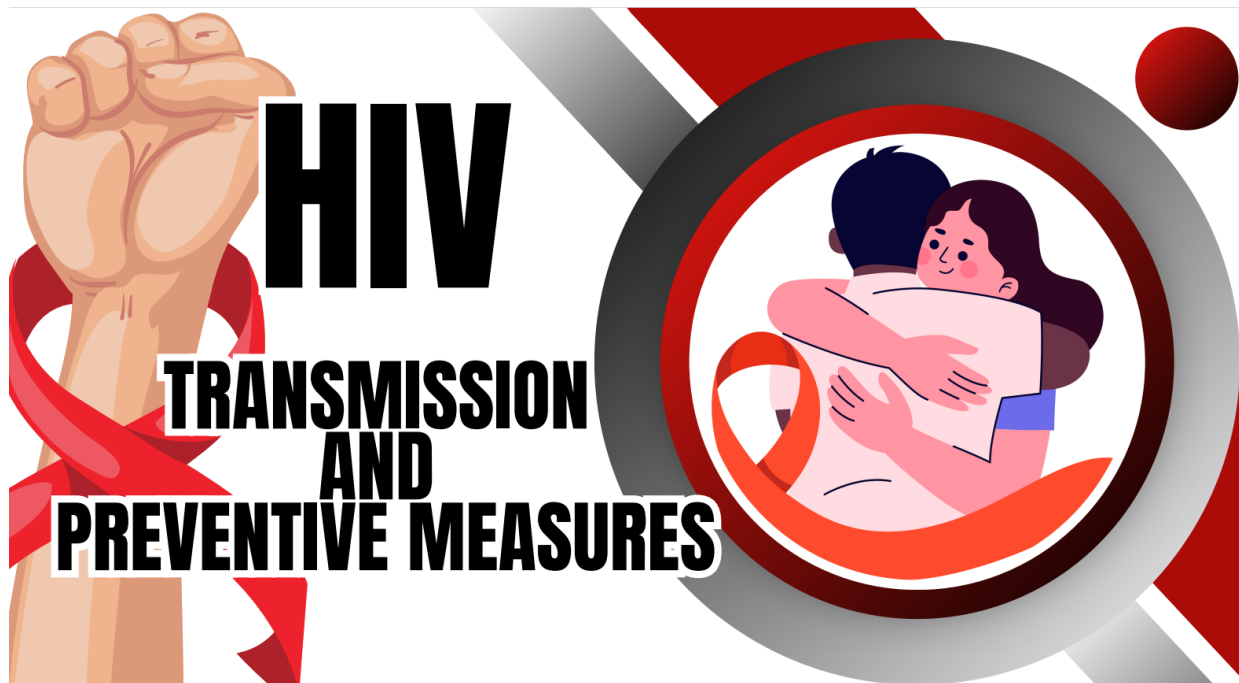


TABLE OF CONTENTS

Title Page	1
Table of Contents	2
HIV and AIDS	3
Mode of Transmission	4-6
Myths about HIV Transmission	7-9
Preventive Measures	10-11
Treatment and Diagnoses	12-13
References	14

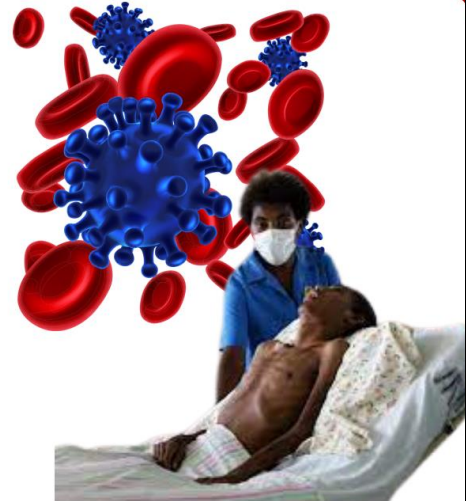


	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 20 of 22

HIV and AIDS

Human immunodeficiency virus (HIV) is an infection that attacks the body's immune system.

Acquired immunodeficiency syndrome (AIDS) is the most advanced stage of the disease.



MODE OF TRANSMISSION



BLOOD

- Blood transmission
- Blood transfusions
- Sharing of needles
- Organ donations
- Needle pricks



	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 21 of 22



MYTHS ABOUT HIV TRANSMISSION





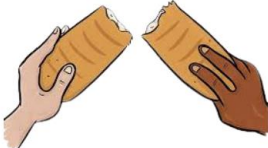
You get HIV through:



HUGGING



KISSING



SHARING OF FOOD



MOSQUITO BITES



MYTHS ABOUT HIV TRANSMISSION





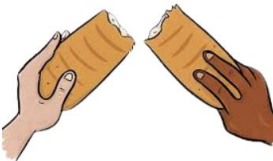
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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 22 of 22

TREATMENT AND DIAGNOSES



IS THERE A CURE FOR HIV?

- **No. There is no cure for HIV as of 2022.**
- **Once you have HIV, you have it for life.**
- **You cannot cure HIV, but you can PREVENT IT FROM GETTING WORSE through medications.**

REFERENCES:

Almeida, C.A. (2022). HIV-Syphilis-and Gonorrhea-lecture [PowerPoint slides].

World Health Organization. (2023, July 13). HIV and AIDS. World Health Organization. <https://www.who.int/news-room/fact-sheets/detail/hiv-aids>

